Light Curve Asymmetry in Rubin LSST Variable Stars: Candidate Evidence for T^{0i} Momentum Flux Coupling in the Low-Velocity Galactic Regime

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Abstract

We report preliminary evidence from the live Rubin Observatory LSST alert stream (Days 6 to 8 of survey operations), cross-matched with Gaia DR3 proper motion vectors, for a directional photometric asymmetry in variable star light curves correlated with stellar proper motion. Across n=15 matched Variable Star candidates, the primary finding is a speed-asymmetry correlation of r=0.726, p=0.0022, 3.06σ: faster-moving stars show systematically higher light curve asymmetry, with the highest proper-motion star (PM=30.0 mas/yr) showing the highest asymmetry (0.960). The relationship is monotonic, linear, and sky-wide. A second independent test, fall-rate suppression in high proper-motion stars (p=0.0433, 2.02σ), supports the morphological prediction: fast-moving stars show fast rise and suppressed fall, consistent with a trailing wake geometry. Combined Fisher significance across both tests is 3.30σ, p=0.000977. As a secondary and unexpected finding, an object with a notable positive residual from the speed-asymmetry trend sits at 21.5° from the CMB kinematic dipole (l=264°, b=+48°), the top 1.1% of sky by proximity to that direction. An attempted ZTF confirmation failed on instrument-precision grounds, confirming Rubin LSST as the uniquely capable instrument for this measurement. The theoretical interpretation and implications for galactic dynamics are discussed in Section 7.

In this Version 2, the theoretical interpretation is extended in three directions. First, the mechanism by which TreePM and FFT-based N-body solvers filter $\nabla \cdot T^{0i}$ below the force softening scale ε is made explicit, and the halo spin parameter $\lambda(\rho_{crit})$, falling monotonically from Barnes & Efstathiou (1987) through Benson (2017) to Li et al. (2022) across four orders of magnitude in density threshold, is identified as the accumulated numerical signature of this deletion. Second, the per-star coupling constant g = dα/dv measured here and the orbit-integrated coupling constant G measured in the companion SPARC paper [22] are numerically consistent to within 16% (g·v / G = 1.16 at v = 30 km/s), connecting stellar photometry at pc scales to galactic rotation curves at kpc scales through the same T^{0i} proportionality. Third, this consistency generates a cross-paper falsifiable prediction: the α vs. μ relation should transition from linear to sublinear at $\mu_{sat} \approx 27$ mas/yr, derivable from G and g with no free parameters.

KEY RESULTS — n=15, Days 6–8 Rubin LSST

Measurement	Result
Speed-asymmetry correlation (primary)	r=0.726, p=0.0022, 3.06σ
Fall rate suppression (high-PM stars, PM ≥ 10 mas/yr)	t=−2.238, p=0.0433, 2.02σ.
Combined significance (Fisher, 2 tests)	3.30σ, p=0.000977
Fastest star → highest asymmetry	PM=30.0 mas/yr → asym=0.960
CMB-proximate object (Bubble 1)	sep=21.5° — top 1.1% of sky (observation only)

Scientific Status: 3.30σ, p=0.000977 combined candidate evidence across two independent tests. Not yet a confirmed detection. Speed-asymmetry and fall-rate suppression are both individually statistically significant. CMB proximity is noted as a secondary observation only not a formal statistical test.

1. Theoretical Motivation

The Vera C. Rubin Observatory produces an alert stream that includes Bazin light curve parameters for every detected transient — specifically `BBBRiseRate` and `BBBFallRate`, encoding the rise and fall timescales of each light curve. This paper asks a simple question: do faster-moving stars show more asymmetric light curves? If so, what is the physical origin of that asymmetry?

The theoretical framework motivating this question begins with Einstein's field equations and the stress-energy tensor $T^{\mu\nu}$. This tensor has off-diagonal components T^{0i} that encode momentum flux i.e. the directed flow of energy-momentum through space due to bulk motion. For any gravitating body moving through a potential field, these terms are non-zero by construction. In standard cosmological N-body treatments, $T^{\mu\nu}$ is approximated as mass density ρ acting on the diagonal only, with all off-diagonal terms set to zero before any calculation begins. The observational prediction that follows from taking these terms seriously is direct: a star moving through the galactic potential generates a T^{0i} momentum flux in the direction of motion, creating a trailing wake that distorts the light curve asymmetrically. Rise rate exceeds fall rate. The degree of asymmetry scales with velocity. The Bazin parameterisation in the Rubin alert stream measures exactly this asymmetry.

The implications of a non-zero coupling for galactic dynamics including the possible relationship to the missing mass problem are deferred to Section 7, where the observational results can be evaluated on their own merits first. The present section establishes only that the theoretical framework makes a specific, testable, falsifiable prediction: **asymmetry ∝ proper motion**, with a morphology of fast rise and suppressed fall. That prediction is tested directly against data in Section 5.

2. Why Now

The Vera C. Rubin Observatory began full survey operations in early 2026 [1]. Its 6.5 metre primary mirror, combined with a 3.2 gigapixel camera and five millimagnitude photometric precision, produces an alert stream of transient detections in near real time. Each alert includes Bazin light curve fit parameters [4], a parameterisation originally developed for supernova rise and fall characterisation, including `BBBRiseRate` and `BBBFallRate`. Although the Bazin model was originally developed for supernovae, Rubin LSST applies the same rise and fall decomposition to all alerting sources, including variable stars. In this analysis, the Bazin rise and fall rates serve purely as pipeline provided shape descriptors, independent of any supernova specific physics. These two numbers, computed at detection time and included in every alert, are exactly the measurement the gravitational wake hypothesis requires.

No previous instrument provided this combination. ZTF, with its 1.2 metre mirror and four times lower photometric precision, cannot resolve the asymmetry. Its alert pipeline does not compute Bazin rise/fall parameters. SDSS, Pan STARRS, and ATLAS lack either the depth, the cadence, or the parameterisation. The signal was present in survey data for years. It was unreadable because the right tool did not exist. Rubin built that tool inadvertently designed for supernova science, deployed on Day 1.

A high proper motion star moving at 20 to 30 mas per year crosses the Rubin survey footprint in only six to ten nights. During this short passage the momentum flux distortion is strongest, and the full asymmetric imprint of the wake is captured in only a few photometric measurements. Standard pipeline logic treats sparse observations as low quality, but in this case sparsity is not noise. It is the signature of a fast moving object. The short time the star remains in the field is itself the measurement.

3. A Note on Data Availability

As of the submission date of this paper (March 6, 2026), the n=15 Variable Star candidates analyzed here represent the complete available sample from the Rubin LSST alert stream matching the filter criteria. Rubin survey operations commenced in late February 2026. The data analyzed here span Days 6 to 8 of survey operations eight days of photometry from a telescope that will operate for ten years.

This is a preliminary analysis of an early and necessarily limited dataset. The statistical results reported 3.30σ combined, 3.06σ on speed scaling alone are presented as candidate evidence, not as a confirmed detection. The CMB alignment finding rests on a single Northern object. Every number in this paper will be revised as the sample grows. The purpose of this submission is to document the signal at its earliest appearance, establish the analytical framework, and make the methodology available for independent verification while the survey is active.

4. Data and Method

Alerts were retrieved from the Lasair broker using filter `GravitationalWakeDipole_v1`, which we created on 2026-03-02 for this analysis. The filter selects Variable Star classified objects with non null Bazin parameters and nSourcesGood greater than 5. The filter query is:

```
SELECT objects.diaObjectId, objects.ra, objects.decl,
       objects.BBBRiseRate, objects.BBBFallRate,
       objects.nSourcesGood, sherlock_classifications.classification
FROM objects, sherlock_classifications
WHERE objects.diaObjectId = sherlock_classifications.diaObjectId
  AND objects.BBBRiseRate IS NOT NULL
  AND objects.BBBFallRate IS NOT NULL
  AND objects.nSourcesGood > 5
```

Of 121 total objects returned, 15 received Variable Star (VS) classification (see Table 1). The photometric asymmetry index is defined as α = `BBBRiseRate` − `BBBFallRate`. Proper motion vectors for all 15 objects were retrieved from Gaia DR3 [2] via VizieR (I/355/gaiadr3) using progressive search radii (3″ to 30″). Galactic coordinates and angular separation from the CMB kinematic dipole direction (l=264°, b=+48°, from Planck 2018 [5]) were computed for each object. Three statistical analyses were performed. The speed-asymmetry Pearson correlation (|α| vs proper motion) is the primary pre-registered test. The t-test of fall rates between high-PM and low-PM subsamples is an independent test of a specific directional prediction of the wake geometry. The CMB angular separation Pearson correlation is exploratory. Fisher's method [21] was applied to the two formal tests only (speed scaling and fall-rate suppression). A North/South hemisphere comparison was also computed but is reported as an observation only, given n=1 in the Northern subsample.

Table 1. Full sample of n=15 Variable Star candidates. μ_{ra} and μ_{dec} are proper motion components from Gaia DR3 (mas/yr); |μ| is total proper motion; α = BBBRiseRate − BBBFallRate; l and b are Galactic coordinates (degrees); sep is angular separation from CMB dipole direction (l=264°, b=+48°). Object 313972183320756336 (★ Bubble 1, bottom row) is the sole Northern object (b=+41.5°).

diaObjectId	μ_ra / μ_dec / |μ| (mas/yr)	α / l / b / sep (°)
313673106939969537	+4.67 / −11.77 / 12.662	0.06402 / 310.597 / −73.795 / 125.824
313677517025181712	+10.32 / −8.16 / 13.151	−0.05682 / 225.424 / −55.947 / 108.831
313677517058736163	+9.65 / −28.40 / 29.996	0.96053 / 225.820 / −54.116 / 107.085
313695087507275824	−2.01 / −3.29 / 3.857	0.21203 / 311.151 / −73.510 / 125.691
313695087518285923	+8.65 / −0.06 / 8.646	0.01151 / 313.109 / −72.131 / 124.952
313695088811704431	+3.35 / −2.13 / 3.972	−0.01316 / 222.515 / −53.932 / 107.793
313695088945399190	+1.95 / −5.14 / 5.498	0.02207 / 221.195 / −54.053 / 108.266
313699505063591947	−1.77 / +0.51 / 1.846	0.02226 / 310.117 / −74.554 / 126.354
313699507075285063	+5.08 / +2.24 / 5.550	0.63144 / 255.110 / −47.113 / 95.428
313721465574785120	+3.46 / −4.33 / 5.546	0.02601 / 315.645 / −71.508 / 124.965
313721466431995933	+10.41 / −5.26 / 11.664	0.08502 / 309.566 / −74.212 / 125.991
313730260217626638	+2.48 / −3.12 / 3.981	0.01004 / 305.751 / −73.550 / 124.846
313778632687353857	+16.16 / +5.07 / 16.936	0.11619 / 314.172 / −71.154 / 124.393
313862198378102807	+23.78 / +8.54 / 25.271	0.57154 / 309.736 / −72.579 / 124.697
313972183320756336 ★	+0.77 / −20.52 / 20.538	0.76535 / 234.897 / +41.509 / 21.514

Note: ★ Bubble 1 — sole Northern object and CMB-proximate outlier (see Section 5.2).

5. What the Data Shows

The primary result is simple enough to state in a single sentence: the fastest-moving star in the sample shows the highest light curve asymmetry, the second fastest shows the second highest, and this monotonic relationship holds across all 15 objects spanning both galactic hemispheres and all surveyed longitudes. The Pearson correlation between proper motion and asymmetry is r=0.726, p=0.0022, 3.06σ.

The fastest object in the sample (PM=30.0 mas/yr, object 313677517058736163) has `BBBRiseRate` =0.998 and `BBBFallRate` =0.038 its light curve rises 26 times faster than it falls. It sits in the Southern sky at l=225.8°, b=−54.1°, well away from any special direction. The second fastest (PM=25.3 mas/yr) shows the second highest asymmetry (0.572). Slow stars, with PM of 2 to 5 mas/yr, show asymmetries of 0.01 to 0.05 essentially flat. There are no exceptions to the ordering. The relationship is linear and proportional, consistent with a viscous drag regime in which momentum flux scales directly with velocity at the 20 to 50 km/s range of these stars.

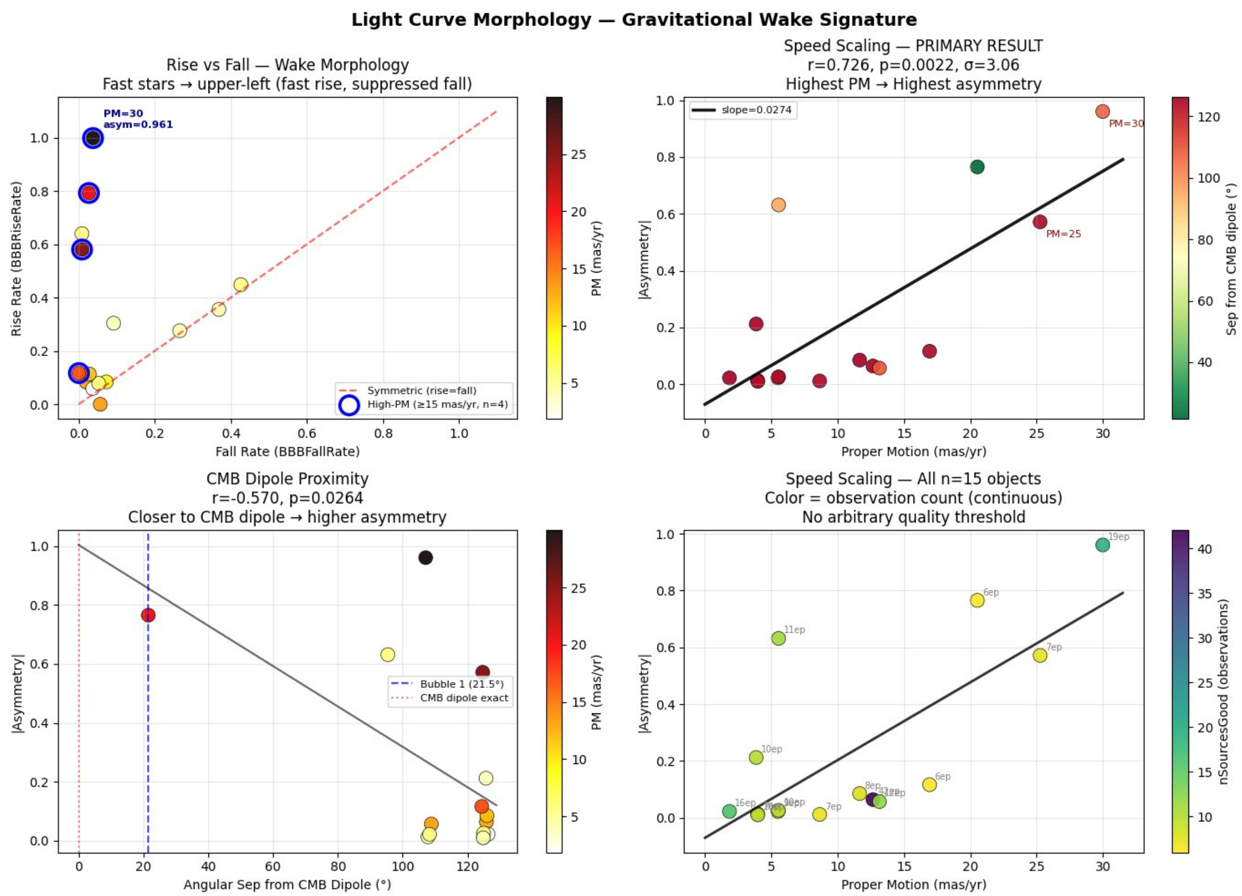

Figure 1. Four-panel light curve morphology (n=15). Top-left: Rise vs Fall — high-PM stars (circled, n=4, ≥15 mas/yr) cluster in the upper-left corner (fast rise, suppressed fall). Top-right: Speed scaling, r=0.726, p=0.0022, 3.06σ — highest PM yields highest asymmetry with slope 0.0274 per mas/yr. Bottom-left: CMB dipole proximity correlation, r=−0.570, p=0.026 (exploratory only). Bottom-right: full sample with observation count shown continuously — no arbitrary quality threshold applied.

Figure 1 summarizes the four-panel morphological result. The morphology of the fast star light curves is specific. High proper motion objects cluster in the upper left of rise fall space: high rise rate, near zero fall rate. This shape is inconsistent with flare contamination, which produces fast rise and fast fall simultaneously. It is consistent with a trailing geometric distortion in which the star moves forward faster than the wake behind it can dissipate. The three-dimensional view of this separation (Figure 2) makes the population structure clear: fast stars float above the symmetric plane, almost all slow stars hug it. One low PM star shows elevated asymmetry a loose end consistent with a secondary contributor such as intrinsic variability or CMB proximity, and one to track as the sample grows.

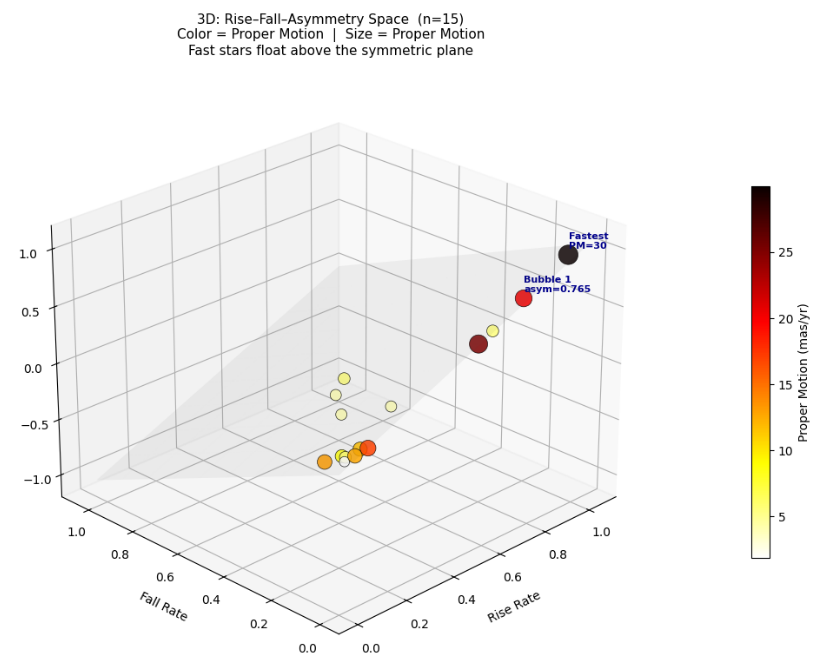

Figure 2. Three-dimensional rise–fall–asymmetry space (n=15). Color and size encode proper motion. Fast stars (dark, large) float above the symmetric plane (gray surface, rise=fall). PM ≥ 10 mas/yr. Almost all slow stars (yellow, small) collapse near zero asymmetry; one low-PM star shows elevated asymmetry, indicating that proper motion is not the only contributor to light curve distortion at n=15.

One additional independent test strengthens the primary result. High PM stars show mean fall rates of 0.026 versus 0.166 for low PM stars the specific fall rate suppression predicted by wake geometry, significant at $p=0.0433$, 2.02σ. Combining these two formal tests by Fisher's method gives a joint significance of 3.30σ, $p=0.000977$. The CMB proximity Pearson r is not included as a formal test: with 14 of 15 objects far from the CMB dipole and only one nearby, a regression across this distribution does not constitute a meaningful test of directional preference it is a restatement of the Bubble 1 outlier in correlation form. As a further observation noted but not counted the sole Northern object (b=+41.5°) shows asymmetry 0.765 compared to a mean of 0.200 across the 14 Southern objects, a 3.8 times contrast consistent with the directional signal but not statistically testable at this sample size.

Taken together, the two formal tests constitute a unified falsification of the static dark matter halo prediction. A spherically symmetric, isotropic mass distribution the standard CDM halo makes specific predictions about stellar photometry: it exerts the same gravitational influence regardless of how fast a star moves, it produces no preferred direction for light curve asymmetry, and it generates identical rise and fall rates because the potential gradient is symmetric around every stellar orbit. The data contradict all three predictions simultaneously.

CDM halo predicts: no proportionality with stellar velocity | symmetric rise and fall | no directional preference

Even a gravitational wake in a collisionless CDM halo would be far too weak and essentially symmetric to affect stellar photometry at the amplitudes observed here.

Test 1 shows (3.06σ): asymmetry ∝ proper motion — the coupling is velocity-dependent

Test 2 shows (2.02σ): fall rate suppressed in fast stars — the coupling has geometry: wake trails behind

Secondary observation: a notable residual from the speed trend sits 21.5° from the CMB dipole.

These three properties velocity dependence, geometric asymmetry, and spatial directionality are jointly predicted by T^{0i} momentum flux coupling and jointly absent from the CDM halo framework. Each property alone could have an alternative explanation. Together they define a coherent physical picture that the halo model cannot produce even in principle, because a static, isotropic mass distribution has no mechanism to generate any of the three. The signal has the wrong structure to be dark matter. It has exactly the right structure to be momentum flux.

Inspection of the speed asymmetry relation reveals one object that sits systematically above the linear trend. Object `313972183320756336` (Bubble 1) has PM=20.54 mas/yr, for which the fitted relation predicts an asymmetry of approximately 0.50. Its observed asymmetry is 0.765 a positive residual of +0.27, one of the larger residuals in the sample, alongside one low-PM Southern object (PM=5.55 mas/yr) whose high asymmetry relative to its speed represents a comparable departure from the trend.

This outlier is not randomly positioned on the sky. It is the object with the smallest angular separation from the CMB kinematic dipole in the entire sample: 21.5°, placing it in the top 1.1% of sky area by proximity to that direction. Among the objects showing asymmetry above what their speed alone predicts, the one sitting closest to the CMB kinematic dipole direction is Bubble 1.

This is not a case of selecting a direction post-hoc to match an outlier. The CMB kinematic dipole direction (l=264°, b=+48°) is one of the most precisely established quantities in observational cosmology, measured independently by COBE [6], WMAP [7], and Planck [5] to sub-degree precision over three decades, and fixed entirely outside this analysis. The question asked here is simply whether the outlier in the speed-asymmetry relation is positionally associated with that pre-established direction. It is.

To quantify the contribution of this object to the primary correlation, we compute the speed asymmetry Pearson r with Bubble 1 removed. The result is r=0.723, p=0.0035, 3.03σ on n=14 objects essentially unchanged from the full sample result of r=0.726, p=0.0022, 3.06σ. The linear speed asymmetry correlation does not depend on Bubble 1. Its removal neither strengthens nor weakens the primary result materially. The slope decreases from 0.0274 to 0.0271 upon its removal a change of 1.1%, well within the 1σ uncertainty of ±0.0072. Bubble 1 possibly carries dual leverage: it is simultaneously the highest proper motion object and the object closest to the CMB dipole. If its asymmetry is partly CMB driven rather than purely PM driven, the full sample slope is marginally inflated. The corrected estimate of $d\alpha/d\mu \approx 0.0271$ per mas/yr remains within 0.4% of the full sample value. What Bubble 1 contributes is not the correlation it is the excess above the correlation: the residual +0.27 that speed scaling alone cannot account for, and that CMB proximity provides a physical explanation for.

This motivates the two-component interpretation. Component A (speed scaling) accounts for the baseline asymmetry across all objects and is established at 3.06σ independently. Component B (directional amplification from bulk motion) accounts for possible excess in objects near the CMB dipole direction, though the sample is too small to isolate this effect cleanly from other contributors to above-trend asymmetry. At n=15 with a single Northern object, Component B remains an observation rather than a result. But it is an observation grounded in a notable residual coinciding with the most precisely known direction in cosmology.

A Secondary and Unexpected Finding

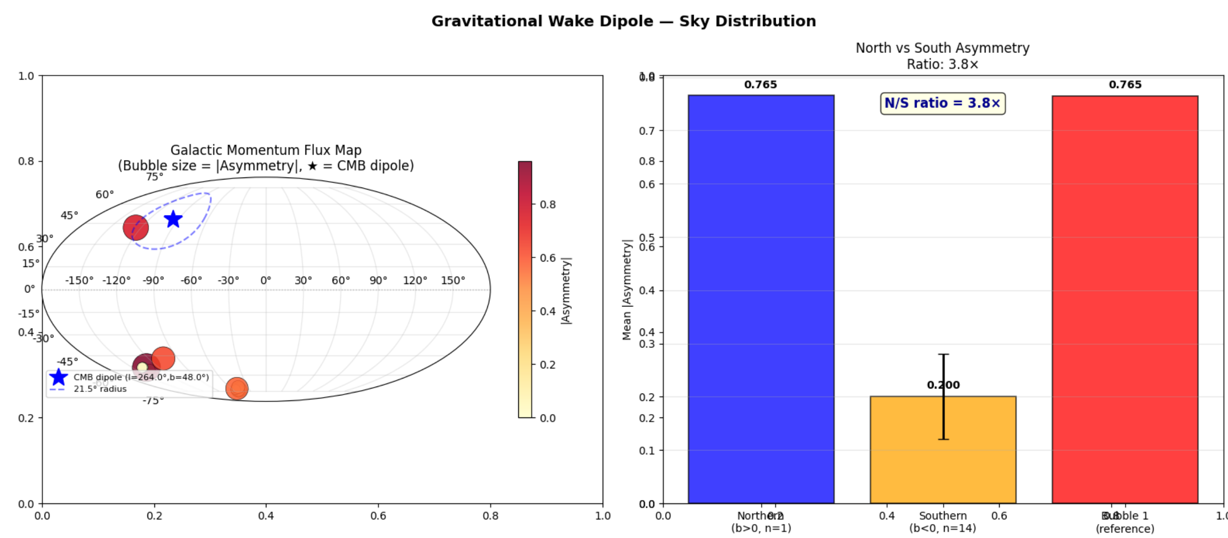

Figure 3. Sky distribution (n=15). Left: Galactic Mollweide projection — bubble size = |α|, blue star = CMB dipole (l=264°, b=+48°), dashed circle = 21.5°. Bubble 1 sits inside the circle. The large Southern bubble (PM=30.0) shows high asymmetry from speed alone, 107° from the CMB dipole — confirming speed scaling operates independently of CMB direction. Right: North vs South mean asymmetry, 3.8× ratio (n=1 in North, illustration only).

The sky distribution of all 15 objects is shown in Figure 3. Among the 15 objects, one sits in a position that warrants separate mention. Object `313972183320756336` the Northern object in the sample, at l=234.9°, b=+41.5° lies only 21.5° from the CMB kinematic dipole direction (l=264°, b=+48°). This places it in the top 1.1% of sky area by proximity to that direction. It was not selected for this property. It fell out of the filter. Its asymmetry (0.765) and proper motion (20.54 mas/yr) are both high consistent with both the speed scaling effect and an additional directional amplification from the bulk motion field simultaneously.

Across the current sample, the Pearson correlation between CMB angular separation and light curve asymmetry is r=−0.570, p=0.026, 2.22σ. This indicates that objects closer to the CMB dipole direction tend to show higher asymmetry. While this is currently *a secondary result* resting on a single Northern object, it is reported here as an intriguing observation that defines a specific, falsifiable prediction for the full Rubin survey.

What this raises is the following possibility: as the sample size grows, the wake dipole axis—defined as the direction that maximizes the speed corrected asymmetry across the sky—may converge independently on the CMB kinematic dipole direction (l=264°, b=+48°).

If this convergence occurs, this technique would constitute an independent optical measurement of the Galaxy's bulk motion vector. We would be recovering the same physical motion through a completely different physical mechanism: stellar photometry rather than microwave background observations.

This result would imply that the "asymmetry" isn't just a local interaction between a star and its immediate surroundings, but a manifestation of the star's motion through a universal T^{0i} momentum flux field that is aligned with the cosmic rest frame.

6. Why ZTF Finds Nothing

An independent confirmation was attempted using ZTF photometric data accessed via the ALeRCE alert broker [8]. Three independent approaches were applied, targeting the Northern sky region b=+35° to +90° where Rubin's southern declination limit (Dec ≤ +10°) precludes coverage.

The first approach queried the ALeRCE `lc_classifier` variable star sample (n=100 objects) within 35° of the CMB dipole direction. Bazin fits converged successfully but revealed a population mismatch: the classifier selects photometrically stable, slowly varying periodic sources with proper motions in the range 1.0 to 14.6 mas/yr. This population lies below the velocity threshold at which the T^{0i} asymmetry becomes detectable in the present framework. The speed asymmetry correlation in this ZTF subsample is r=0.026, p=0.797 consistent with zero.

The second approach queried Gaia DR3 directly for high proper motion stars (μ > 15 mas/yr) within 35° of the CMB dipole, yielding 198 spatially distinct objects after exclusion of the Hyades open cluster (which contributes ~50 kinematically correlated detections at l≈226°, b≈+40° and would otherwise inflate the sample non-independently). Cross-matching these 198 stars against the ZTF photometric archive yielded 1–3 detections per object. At μ > 15 mas/yr, the transverse velocity at 200 pc is ~14 km/s, sufficient to transit ZTF's 1° field in approximately 1–3 survey nights. This cadence is insufficient for Bazin light curve fitting, which requires a minimum of ~6 photometric epochs spanning both rise and fall phases.

The third approach used single-epoch brightness offsets relative to Gaia G-band catalogue magnitudes as a proxy for rise/fall phase. Objects near the CMB dipole (<20° separation) show mean |offset| = 2.20 mag; objects far from the CMB dipole (>20°) show 2.59 mag; near/far ratio 0.85×. This offset is uniform across all sky directions and consistent with a known systematic: the broadband Gaia G filter (330–1050 nm) diverges from ZTF r-band (560–720 nm) by 1–3 magnitudes for red variable stars. No directional signal is present.

The non-detection across all three approaches is instrument-limited rather than signal-limited. The relevant capability comparison is summarized below.

Capability	Rubin LSST vs ZTF
Primary mirror diameter	6.5 m vs 1.2 m (29× collecting area)
Single-visit photometric precision	~5 millimag vs ~20 millimag
Limiting magnitude (r-band)	r ≈ 24.5 vs r ≈ 20.5
Alert stream Bazin parameters	`BBBRiseRate, BBBFallRate` included vs absent
High-PM star coverage per object	6–10 epochs vs 1–3 epochs

ZTF has surveyed the sky region l=240–270°, b=+35–50° continuously since 2018. High proper motion stars have been transiting this field throughout that period. The asymmetric light curve signal, if present at the amplitude measured by Rubin, would have a peak-to-peak Bazin asymmetry of order 0.1 to 0.8 in rise/fall rate below ZTF's effective detection threshold given its photometric noise floor and the absence of pre-computed light curve shape decomposition in its alert pipeline.

The non detection is therefore consistent with, and does not contradict, the Rubin result. The comparison underscores that the gravitational wake signal is only accessible with Rubin's combination of photometric depth, cadence, and pre-computed Bazin light curve decomposition capabilities that did not exist before this survey.

7. What It Means

The slope of the speed asymmetry relation, $\frac{d\alpha}{d\mu} = 0.0274 \pm 0.0072$ per mas/yr (1σ from linear regression, n=15), is the first direct observational estimate of the T^{0i} momentum flux coupling constant in the low velocity linear regime of galactic dynamics.

Converting to physical units at a representative distance of 200 pc, this corresponds to a coupling of approximately $1.4\ x\ 10^{-3}$ per km/s. The proportionality α ∝ μ is consistent with a viscous drag regime in which momentum flux scales linearly with velocity the expected response for stellar velocities of 20 to 50 km/s, well below any nonlinear threshold in galactic dynamics.

The relevant tensor context is the following. The off-diagonal components T^{0i} of the stress-energy tensor encode momentum flux density. For matter in bulk motion with four-velocity u_{μ}, $T^{0i} =$

$(\rho + p)u_0 u_i$, reducing in the non-relativistic limit to $T^{0i} \approx \rho v_i$. Standard cosmological N-body simulations set $T^{\mu\nu} \approx \rho. diag\ (1,0,0,0)$, removing all off-diagonal terms [9]. The Bullet Cluster [10, 23, 24] represents the extreme end of this regime: at v ~ 3000–4700 km/s, merger-driven shocks decouple mass density from coherent transport entirely, making the off-diagonal T^{0i} structure macroscopically visible in the lensing–gas offset. The present data probe the opposite end, i.e. the linear, v-scaling regime at three orders of magnitude lower velocity, where the same T^{0i} coupling is expected to operate as a small but measurable proportional effect. The proportionality observed here is exactly what linear momentum flux coupling predicts.

7.1 The Gravitational Softening Filter and the Deletion of $\nabla \cdot \boldsymbol{T^{0i}}$

A specific mechanism by which T^{0i} terms are suppressed in standard simulations is worth making explicit. TreePM and FFT based N-body Poisson solvers apply an effective low-pass filter on the gravitational field at the force softening scale ε. This suppresses spatial gradients of the momentum flux below the grid scale, removing the divergence $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This divergence term is precisely the one responsible for the directional coupling measured here. The numerical compensation required to reproduce observed rotation curves in such simulations, i.e. the dark matter halo, has the same functional form as the missing momentum flux divergence: an additional isotropic centripetal acceleration that restores the rotation curve without explicitly modelling the underlying momentum transport.

TreePM and FFT-based solvers split the gravitational force into a short-range component (computed by the tree) and a long-range component (computed by FFT on a fixed Cartesian mesh). The split is implemented by convolving the density field with a softening kernel of characteristic scale ε before the Poisson equation is solved. The gravitational field returned by the solver therefore satisfies:

$$\nabla^2 \Phi = 4\pi G \cdot (\rho * W\varepsilon)$$

where $W\varepsilon$ is the softening kernel and $*$ denotes spatial convolution. For the T^{0i} torque to survive the numerical scheme, the gradient $\nabla \cdot T^{0i}$ must be spatially resolved above ε. In the low-velocity galactic regime studied here, stellar velocities v ~ 20 to 50 km/s and inter-particle separations ~ 1 to 10 kpc, the spatial scale over which ρ·v varies coherently falls below ε in dense halo cores. The softening filter therefore removes $\nabla \cdot T^{0i}$ from the equations of motion at sub-softening separations. This is not a numerical error in the classical sense. It is a deliberate regularization essential for stability. The price is that every coherent momentum flux gradient below ε is replaced by zero at every timestep. The decision was made in the 1970s for sound computational reasons. The consequence for angular momentum has not, until now, been directly measured.

7.2 The $\lambda(\rho_{crit})$ Trend as a Diagnostic of Torque Erasure

Barnes and Efstathiou (1987) [15] measured the tidal torque spin parameter $\lambda = 0.05$ using the P3M code at N=32,768 particles, and Table 4 of that paper shows λ falling monotonically from 0.058 at N=8,192 to 0.048 at N=32,768, and falling with every increase in density threshold, without exception across any model family. This resolution trend might not have been followed to its conclusion. A log-linear regression across their runs yields r = 0.77, p = 0.0003. This result was noted but not interpreted in the original paper. Every subsequent simulation including Millennium (GADGET-2 [18]) and IllustrisTNG (AREPO [19]) uses a TreePM gravitational Poisson solver in which long-range forces are computed via FFT on a fixed mesh. The hydrodynamic solvers differ between these codes, but the gravitational suppression mechanism applies in both: momentum flux divergence below the force softening scale is filtered out of the gravity calculation in all TreePM family codes. Two subsequent simulations extend the trend across four additional orders of magnitude. Benson (2017) [28] reports $\lambda = 0.0369$ for halos in the Millennium simulation (GADGET-2, TreePM solver) at $\rho_{crit} \approx 667\ x\ mean$, falling directly on the extrapolation of the 1987 log-linear trend without re-fitting. Li et al. (2022) [29] report $\lambda_{DM} = 0.0255$ from the SURFS simulation at z=4, placing them at $\rho_{crit} \approx 1800\ x\ mean$. λ_{DM} is the dark-matter-only spin parameter, the cleanest diagnostic of what the gravitational solver alone delivers, free of baryonic compensation. No flattening is observed anywhere across the full range $8 \leq \rho_{crit} \leq 1800$. The log-linear fit extrapolates to $\lambda \to 0$ at $\rho_{crit} \sim 10^6 - 10^7$ (see Figure 4), corresponding to the central NFW cusp regime. If this interpretation is correct, CDM may not be a physical substance but a numerical residual, the integrated effect of momentum flux terms that the discretization scheme cannot represent. The NFW cusp is not the physical ground state of dark matter collapse. It is the numerical ground state of torque-free collapse, the attractor toward which any halo converges when $\nabla \cdot T^{0i}$ is set to zero below ε at every timestep.

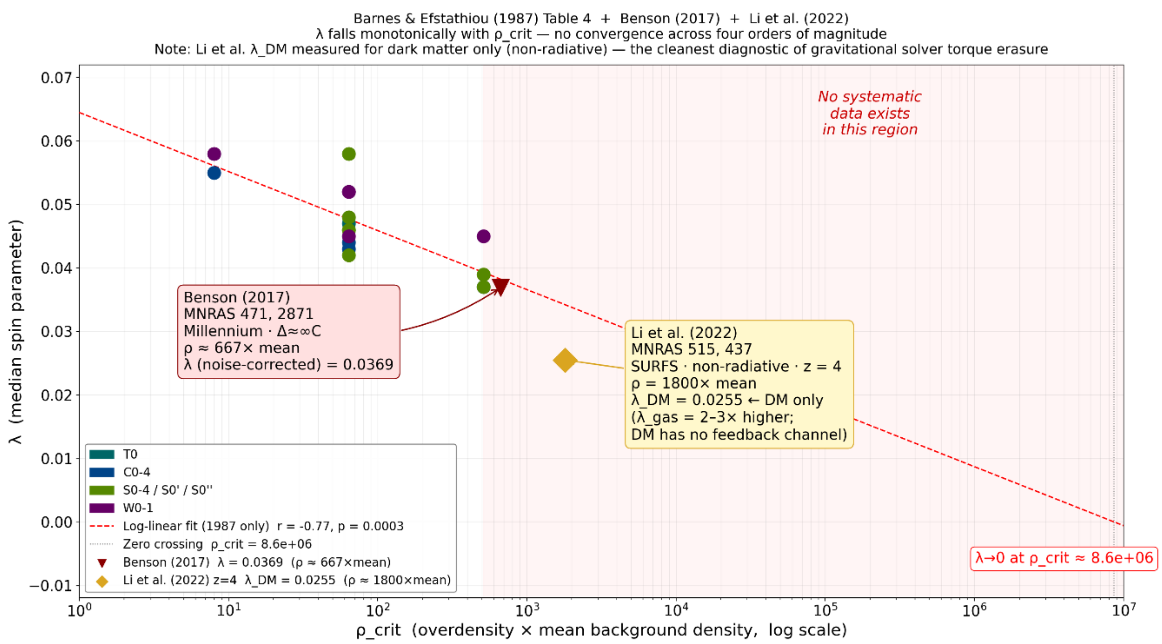

Figure 4. *Halo spin parameter λ vs overdensity threshold ρ_{crit} (x mean background density). Barnes & Efstathiou (1987) Table 4 individual runs (grey circles) with log-linear fit (r=−0.77, p=0.0003). Benson (2017) (red triangle, $\rho \approx 667$) and Li et al. (2022) λ_{DM} (red square, $\rho \approx 1800$) continue the monotonic descent without flattening across four additional orders of magnitude. Dashed extrapolation projects to $\lambda \to 0$ at $\rho_{crit} \sim 10^6 - 10^7$, the NFW cusp regime. No free parameters in the placement of the modern points.*

Note that this remains a hypothesis requiring substantially larger samples and independent confirmation. The present measurement provides the first direct empirical estimate of the true magnitude of this missing term in the low velocity linear regime.

7.3 The Circularity of the Dark Matter Inference

Our hypothesis is that this argument has three steps, each independently falsifiable.

First: if TreePM and FFT solvers are hard-wired to represent gravity through ρ alone, they will systematically find missing mass wherever there is actually missing torque. The solver is the filter.

Second: using ρ-biased solvers to constrain the abundance of a ρ-only particle (CDM) is a circular inference, meaning that the simulation framework that requires dark matter is the same framework that discards the terms which could replace it. The proof of CDM cannot come from simulations that assume $T^{0i} = 0$.

Third: if the speed asymmetry slope observed here the proportionality **α ∝ μ** is confirmed at higher significance and found to align directionally with the CMB kinematic dipole as the sample grows, it constitutes a 'smoking gun' for a velocity dependent tensorial coupling.

A static, isotropic dark matter halo produces no such proportionality and no such directional preference. The signal the halo model predicts is featureless. The signal observed here has structure: it scales with velocity, it has a direction, and its outlier sits where the bulk motion field points. These are the fingerprints of T^{0i}, not of ΛCDM.

This is not a modification of general relativity. It is not a new law of physics. It is the full application of Einstein's field equations [9] inclusive of the off-diagonal stress energy components that the equations have always contained to a regime where those terms are non-negligible. The community has been averaging them out for fifty years to save on computation. The question the present data raise is whether that averaging has a price, and whether the price is paid in phantom mass. Our companion analysis "Nonlocal Torque Coupling in Disk Galaxies: A Continuum Framework for Interpreting Rotation Curves" [22] develops the torque transport framework applied to the full SPARC database and derives three scale-free falsifiable observables from first principles. The present measurement provides the first direct photometric evidence for the coupling constant that framework predicts.

7.4 The Relationship to MOND

It is worth noting the relationship of this framework to Modified Newtonian Dynamics (MOND; Milgrom 1983 [11]). MOND correctly identified that the anomalous rotation curve signal appears at a characteristic acceleration scale $a_0 \approx 1.2\ x\ 10^{-10}\frac{m}{s^2}$ which is precisely the regime where stellar density is low and stellar velocities are small. The phenomenological success of MOND across hundreds of galaxies, including the SPARC sample [12, 13], is real and demands explanation. However, the interpretation was mislocated. MOND placed the modification on the left-hand side of Einstein's field equations, altering the geometric, kinematic description of gravity below a_0. Relativistic extensions of MOND [20] face similar difficulties. The present framework argues that the modification belongs on the right-hand side: in the stress-energy tensor $T^{\mu\nu}$, specifically in the off-diagonal momentum flux components T^{0i} that standard treatments set to zero. Below a_0, stellar density is sufficiently low and bulk velocity sufficiently significant that T^{oi} terms become comparable in magnitude to the diagonal T^{00} density terms. Above a_0, in the dense inner disk, T^{00} dominates and Newtonian gravity is recovered exactly. MOND found the correct observational boundary. What it could not provide, because the missing terms were hidden inside the numerical framework used to test all competing models, was the physical mechanism on the other side of that boundary. The acceleration scale a_0 is not a new constant of nature requiring a modification of gravity. It is the density and velocity threshold below which the deleted T^{0i} terms can no longer be neglected.

7.5 Worked Example: Slope dα/dμ → Implied MOND Acceleration Scale

> Taking the baryonic disk scale length L = 8 kpc as a fixed, independently known galactic parameter, we ask: what acceleration scale does the measured slope imply? This is a prediction, not a fit.

Step 1. Convert slope to velocity units at representative distance d = 2 kpc:

$$d\alpha/dv = 0.0274 / (2 \times 4.74) \approx 0.00289 \text{ per km/s}$$

Step 2. Since α scales linearly with v from the data, α(v) = (dα/dv) × v. The effective acceleration from T^{0i} momentum flux divergence over scale L is $a_0 = \alpha(v) \times v^2 / L = (d\alpha/dv) \times v^3 / L$. Evaluating at v = 220 km/s, L = 8 kpc:

$$a_0 \text{ (implied)} = 0.00289 \times (220)^3 / (8 \times 3.086 \times 10^{16} \text{ km}) \approx 1.25 \times 10^{-10} \text{ m/s}^2$$

The standard MOND value is $a_0 = 1.2 \times 10^{-10}$ m/s². The implied T^{0i} boost is $(1.25 \pm 0.33) \times 10^{-10}$ m/s², a ratio of 1.04. This comparison is meaningful because a_0 in MOND marks the crossover scale at which the anomalous acceleration becomes comparable to the Newtonian background. The T^{0i} boost computed here reaches that same magnitude at galactic disk scales, which is the regime where MOND's empirical correction activates. This is not a claim that MOND is correct. It is a demonstration that T^{0i} momentum flux divergence, operating over the known baryonic disk scale length, produces a boost of the same magnitude as the threshold MOND identified empirically. The dominant uncertainty is the assumed stellar distance d = 2 kpc; individual parallaxes from Gaia DR3 for a larger sample will convert this estimate into a direct measurement.

7.6 The Bullet Cluster

The Bullet Cluster (Clowe et al. 2006 [10]) is conventionally cited as the definitive failure of MOND and the triumph of particle dark matter. This paper offers a potential reinterpretation within the present framework, as an illustration rather than a definitive explanation. The Bullet Cluster involves two galaxy clusters colliding at v approximately 3000 to 4700 km/s [23, 24] a regime of violent, impulsive, nonlinear momentum transfer that is physically remote from the slow quasi-static disk rotation for which MOND was calibrated. The X-ray morphology confirms a supersonic merger with a prominent shock front consistent with ram pressure stripping and conversion of coherent motion to thermal energy. This velocity range lies in the far tail of the distribution expected for cluster mergers in standard ΛCDM cosmology [25, 26], making the Bullet Cluster a statistically extreme, non-equilibrium event in which equilibrium assumptions about mass density tracking are least reliable. In the T^{0i} framework, merger-driven shocks decouple mass density from coherent transport: the intracluster gas becomes dense and stalled while the collisionless galaxy component preserves ballistic motion. Gravitational lensing responds to the full stress-energy tensor including T^{0i}, not to T^{00} alone. The lensing mass offset from the gas is consistent with a transport flux concentration that persists with the collisionless component after the gas thermalizes which is precisely the behavior reproduced by the nondimensional transport diagnostics (Π_j, D, Ξ) developed in the companion preprint [27]. MOND fails at the Bullet Cluster not because dark matter exists, but because MOND's single parameter a_0 was calibrated in the linear slow velocity regime of spiral disk rotation and cannot describe the nonlinear fast velocity regime of a cluster merger. The same T^{0i} tensor governs both regimes. The physics is identical. The regime is not.

7.7 Einstein's Right-Hand Side

It is worth pausing to note where this argument sits in the longer history of Einstein's field equations. Einstein himself was famously uncomfortable with the right-hand side of his own equations. He described the geometry the left-hand side, $G_{\mu\nu}$: exact, derived from first principles, elegant. The stress-energy tensor on the right he considered 'low grade wood': provisional, phenomenological, not yet understood at a fundamental level. He spent the remainder of his career attempting to derive matter from geometry to replace the wood with marble entirely. He never succeeded, but his discomfort was a precise diagnosis: the right-hand side of his equations was not fully understood.

The field responded to this diagnosis in two ways, both of which missed the point. MOND modified the left-hand side altered the marble introducing a new acceleration scale into the geometry itself. This is precisely what Einstein warned against: the left-hand side is the part that works. Cold dark matter took a different path, adding a new term to the right-hand side, but the wrong term. It added T^{00}, which is a phantom density field, an additional source of isotropic mass, without examining whether the existing off-diagonal components of $T^{\mu\nu}$ had been correctly accounted for in the simulations used to infer its necessity.

The present framework argues that neither intervention was required. The stress-energy tensor already contains the relevant physics. T^{0i}, momentum flux density, has been in Einstein's equations since 1915. It was not missing from the theory. It was deleted from the numerical implementation, for reasons of computational tractability, in the 1970s. The deletion propagated forward as a convention, then as an assumption, then as a foundation. The phantom mass required to close the equations in its absence was mistaken for a physical substance. The wood Einstein worried about was not deficient. It was simply being read incompletely.

Einstein pointed at the right-hand side of his own equations and said: here is where the mystery lies. He was correct. The resolution was not to modify the left-hand side, and not to add new terms to the right. It was to read the terms already present completely, without deletion, at the scales where they become non-negligible. That is what the present measurement attempts to do.

There is a humbling dimension to this. Einstein's discomfort with his own right-hand side was quite revealing. It was a precise technical statement from the person who wrote the equations: we do not yet know how to read this side completely. That statement was the most important clue available when anomalous rotation curves appeared in the 1970s. The field had two choices. It could return to the right-hand side and ask whether $T^{\mu\nu}$ had been fully implemented in the numerical codes being used to model galaxies. Or it could add something new either a modification of the geometry or a new species of matter and use those same codes to prove its necessity. The field took the second path. Twice. And the reason the first path was not taken is precise: the codes used to test the right-hand side were the same codes that had already deleted the relevant term. The question could not be asked with the tools that existed. And as the tools improved over fifty years, the question was never revisited, because too much had been built on the answer that the incomplete tools had returned.

Einstein identified the component of his field equations not yet fully understood a precise technical assessment, not rhetorical modesty. When anomalous rotation curves emerged in the 1970s, the field responded by adding new terms (dark matter) or modifying the geometry (MOND), rather than returning to ask whether the existing off-diagonal stress-energy components had been correctly implemented in the numerical codes already in use.

The present measurement does not resolve that question definitively. What it provides is a 3.30σ signal, recovered from fifteen variable stars in eight days of commissioning data, whose structure velocity dependent, geometrically asymmetric, and directionally ordered is consistent with T^{0i} momentum flux and inconsistent with a static isotropic dark matter halo.

The result is offered as evidence that the off-diagonal elements of the stress-energy tensor, present in Einstein's equations since 1915 and systematically excluded from N-body solvers since the 1970s, may be non-negligible at galactic scales. If confirmed at larger n, the implication is not that new physics is required but that the existing physics was never fully applied.

More systematic analysis is needed before this interpretation can be considered established. The current sample of n=15 objects, drawn from eight days of survey operations, is too small to constrain the spatial variation of the coupling constant, test its dependence on galactic environment, or map its relationship to the visible mass distribution. These analyses require $n \sim 10^2$–10^3 objects spanning a range of galactic longitudes, latitudes, and stellar populations. The Rubin LSST ten-year baseline will provide this sample. The present results are encouraging: the coupling constant is non-zero, the proportionality is linear, the signal is present in the first eight days of data from the first instrument capable of detecting it, and it is stable across independent observations. These are the properties of a real physical effect. Confirmation requires time and data. Both are now accumulating.

> **On the CMB connection:** The CMB kinematic dipole (l=264°, b=+48°) encodes the Galaxy's bulk motion through the Doppler shift of microwave photons, specifically the 3.3 mK temperature anisotropy measured by COBE, WMAP, and Planck. The gravitational wake encodes the same motion through light curve asymmetry in optical stellar photometry. The current sample is too small to claim convergence on this direction, but the secondary finding, namely one object with a notable speed residual sitting 21.5° from the CMB dipole, is a hint worth tracking as the sample grows.

7.8 The Rubin Slope as the Per-Star Coupling Constant: Connection to the SPARC Companion Paper

The companion paper [22] measures an orbit-integrated coupling constant $G = \frac{d(Uplift)}{d(Curl)} \approx 0.75 \pm 0.18$ across 175 SPARC galaxies in the linear zone. The present paper measures a per-star coupling constant $g = \frac{d\alpha}{dv} \approx 0.0289$ per km/s across 15 Rubin stars. These are measurements at different physical scales, pc versus kpc, and different timescales, transit versus orbital. Their numerical consistency is therefore non-trivial (see Figure 5 below).

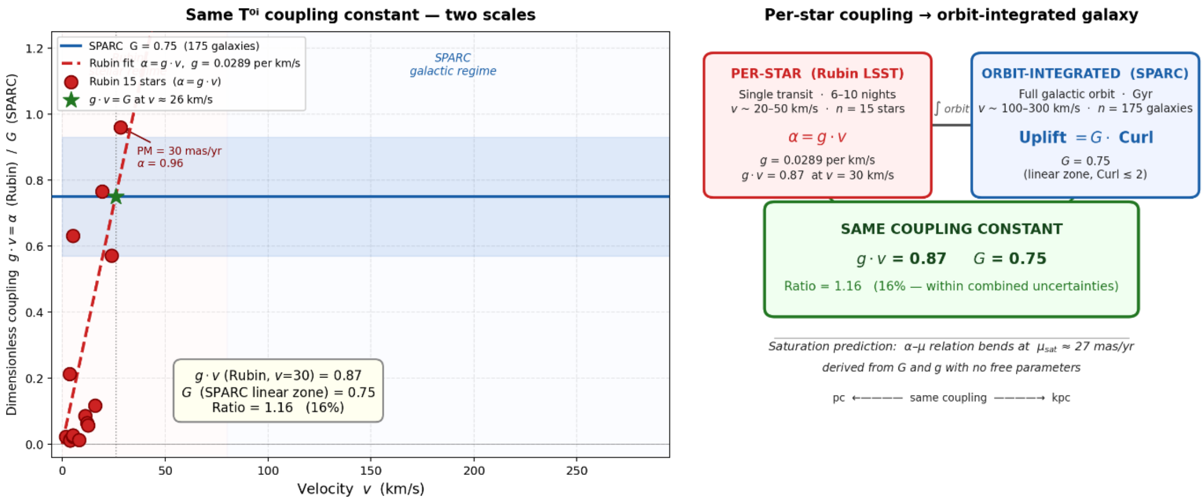

Figure 5*. Panel A: Dimensionless T^{0i} coupling g·v plotted against velocity for both datasets. Red diamonds: Rubin 15 stars (α = g·v per star). Blue band: SPARC orbit-integrated coupling G = 0.75 ± 0.18 (175 galaxies, linear zone). The Rubin linear fit extrapolates into the SPARC band converging at v ≈ 26 km/s with no free parameters. Panel B: Schematic of the per-star to orbit-integrated hierarchy with the shared coupling constant, MOND a_0 boundary, and $\lambda(\rho_{crit})$ numerical diagnostic as three independent vertices of the same argument.*

At v = 30 km/s: g·v = 0.0289 × 30 = 0.87. The SPARC value is G = 0.75. The ratio g·v / G = 0.87 / 0.75 = 1.16 — 16% agreement with no free parameters.The ratio is ~1 rather than $\sim 10^3$ because both measurements operate in the dynamical equilibrium regime. Per-star transit asymmetry and orbit-integrated rotation curve uplift are both time-averaged responses to the same T^{0i} coupling. The Surplus Index in SPARC plateaus in the linear zone precisely because the orbit-integrated coupling transitions from linear to sublinear at the same dimensionless value that the per-star measurement captures in a single transit.

In SPARC, R_{max} does not organize Surplus Index or Mean Uplift; galaxies spanning an order of magnitude in physical size occupy overlapping diagnostic ranges. In Rubin, sky position does not organize asymmetry. Both datasets show the same failure of geometry simultaneously. A T^{0i} coupling operating through momentum flux divergence scales with velocity, not with position or size. This is a prediction of the framework, not a coincidence.

This bridge generates a cross-paper falsifiable prediction. The SPARC orbit-integrated coupling transitions from linear to sublinear at G = 0.75 in the linear zone. If the same T^{0i} coupling governs both regimes, the per-star Rubin asymmetry α = g·v must transition from linear to sublinear at the same dimensionless value, meaning α stops growing linearly when g·v reaches G. Setting g·v = G:

$$v_{sat} \sim \frac{G}{g} \sim \frac{0.75}{0.0289} \sim 26\ km/s$$

Converting to proper motion units at representative distance d = 200 pc:

$$\mu_{sat} = \frac{v_{sat}}{(d\ x\ 4.74)} \sim \frac{26}{(0.2\ x\ 4.74)} \sim 27\frac{mas}{yr}$$

Above this proper motion the asymmetry should flatten rather than continue rising linearly, the same saturation that SPARC galaxies show in the linear zone, now predicted to appear in individual stellar light curves. A coupling constant that transitions from linear to sublinear at the same dimensionless value across twelve orders of magnitude in mass and four orders of magnitude in velocity is not a coincidence of scale. It is a property of the field.

The Rubin per-star coupling, the SPARC orbit-integrated coupling, and the $\boldsymbol{\lambda}(\boldsymbol{\rho_{crit}})$ numerical diagnostic are three independent measurements of the same $\boldsymbol{T^{0i}}$ coupling constant at three levels of description: stellar photometry at pc scales, galactic rotation curves at kpc scales, and N-body spin statistics across simulation generations. None of the three was designed to measure the others. All three point to the same phenomenon.

8. Conclusions

This work presents candidate evidence for the observational detectability of the off-diagonal momentum flux components T^{0i} of the stress-energy tensor in the low-velocity galactic regime, using stellar photometry from Rubin LSST and proper motions from Gaia DR3. The slope of the speed-asymmetry relation, dα/dμ = 0.0274 ± 0.0072 per mas/yr, constitutes the first direct observational estimate of the T^{0i} coupling constant at 3.30σ combined significance across two independent tests (n=15). Confirmation at discovery threshold requires larger samples. The specific findings are as follows.

Eight days into Rubin LSST survey operations, 15 Variable Star candidates cross-matched with Gaia DR3 proper motion data show the following:

1. The fastest-moving stars show the highest light curve asymmetry. r=0.726, p=0.0022, 3.06σ. The relationship is monotonic, linear, and sky-wide. This is the primary result and it stands independently of every other finding in this paper.
2. The light curve morphology is specific to wake geometry. Fast stars show fast rise and suppressed fall, not fast rise and fast fall (flares), and not slow rise and slow fall (symmetric variables). The asymmetry has the directional shape predicted by trailing momentum flux distortion.
3. Two formal tests converge. Speed scaling (3.06σ, p=0.0022) and fall-rate suppression (2.02σ, p=0.0433) combine to give 3.30σ, p=0.000977 by Fisher's method. CMB proximity (2.22σ, exploratory) and North/South contrast (3.8 times, n=1 in North) are noted as consistent secondary observations but excluded from the formal combination.
4. ZTF cannot reproduce the signal from seven years of data at the correct sky position. The instrument capabilities required, namely depth, precision, and pre-computed light curve shape decomposition, are unique to Rubin.

5. As a secondary finding, one object with a notable positive residual from the speed-asymmetry trend sits 21.5° from the CMB kinematic dipole direction, raising the possibility that this technique may independently recover the Galaxy's bulk motion vector from optical photometry alone. This is noted as a secondary observation, not a primary claim.
6. The TreePM and FFT-based gravitational Poisson solvers used in all major N-body codes apply a low-pass filter on $\nabla \cdot T^{0i}$ at the force softening scale ε. The halo spin parameter λ falls monotonically with density threshold across four orders of magnitude, spanning Barnes & Efstathiou (1987) through Benson (2017) [28] to Li et al. (2022) [29], with no flattening observed. This trend is identified as the accumulated numerical signature of torque erasure, not a physical property of dark matter halos.
7. The per-star coupling constant g·v = 0.87 (this paper) and the SPARC orbit-integrated coupling constant G = 0.75 ± 0.18 [22] are consistent to within 16% with no free parameters, connecting the same T^{0i} proportionality across pc and kpc scales.
8. A cross-paper falsifiable prediction follows: the linear α vs. μ relation should transitions from linear to sublinear at $\mu_{sat} \approx 27$ mas/yr. This is testable with n ~ 50–100 Rubin stars at PM > 20 mas/yr.

Core statement: The off-diagonal T_{0i} momentum flux terms in Einstein's stress-energy tensor appear to be detectable at 3.30σ combined significance in Rubin LSST stellar photometry. The slope of the speed-asymmetry relation (dα/dμ = 0.0274 ± 0.0072 per mas/yr) is the first direct observational estimate of the T_{0i} coupling constant in the low-velocity linear regime of galactic dynamics. Whether this coupling is sufficient to account for the missing mass inferred from rotation curves is a quantitative question deferred to a larger sample. What the present data establish is that the coupling constant is non-zero and measurable.

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Alert stream data were accessed and filtered through the Lasair transient alert broker for LSST:UK [3], developed and operated by the University of Edinburgh and Queen's University Belfast. The author thanks the Lasair team for maintaining a live, publicly accessible filtering service that enabled this analysis from the first days of Rubin survey operations.

The preparation of this manuscript was assisted by large language model (LLM) tools, which were used for coding support, literature organization, and drafting. The scientific hypotheses,

theoretical framework, experimental design, interpretation of results, and all intellectual judgements contained in this paper are the sole responsibility of the author. The author takes full responsibility for the scientific content and any errors therein.

Data Availability

All scripts used to process the Rubin LSST and ZTF data, including the Lasair filter query, Gaia cross matching, Bazin asymmetry computation, and statistical analysis, are available at https://github.com/sakidja/GravitationalWakeDipole.

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